

# Pouyan Navard

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## About Me

Generative AI researcher building efficient, controllable diffusion models for image, video, and 3D generation.

## Skills

**Generative AI & Diffusion Models:** Stable Diffusion, ControlNet, LoRA/QLoRA/PEFT, DPO for Generative Models, Image/Video/3D Generation, Diffusion-Based Animation, 3D Morphable Models (FLAME)

**Vision & Multimodal AI:** Vision Transformers, Vision-Language Models (LLaVA), Masked Multimodal Modeling, 3D Perception, Point Cloud Processing

**Frameworks & Infrastructure:** PyTorch, HuggingFace, PyTorch Lightning, CUDA, DDP/FSDP/DeepSpeed, Hydra, OpenCV, Open3D, SLURM, AWS, Weights & Biases, Git

**Languages:** Python, C++, Bash

## Industry Experience

### Computer Vision Engineer

Path Robotics Inc. [🔗](#)

Columbus, Ohio

Nov 2024 – May 2026

- Designed a **mesh diffusion pipeline** for synthetic 3D asset generation with controllable lighting, texture, and geometry, producing photorealistic training assets at ~2 min/asset on H100.
- Eliminated dependency on manual annotation (\$15/hr external vendor) through diffusion-based synthetic data, reducing data acquisition cost and cutting real-world data collection time by **86.7%**.
- Architected a ~800M-parameter multimodal foundation model trained via cross-modal masked modeling on paired point cloud and image data, consolidating 5 per-customer models into a single generalizable model with **96%** classification accuracy.

## Selected Publications

### Loki: Representation over Architecture for Diffusion-Based Portrait Animation [🔗](#)

Pouyan Navard, Sernam Lim

*Conference on Neural Information Processing Systems (NeurIPS), 2026 (submitted)*

- Developed a diffusion-based portrait animation framework that conditions on FLAME parametric face representations instead of RGB, achieving cross-identity reenactment without cross-ID training data while requiring **43% fewer** parameters and **1,496× less** training video than the leading diffusion baseline.
- Introduced two evaluation metrics (Head Pose Follow, Expression Follow) grounded in the parametric face model, exposing systematic gaps in the field's inherited metric suite.

### KnobGen: Controlling the Sophistication of Artwork in Sketch-Based Diffusion Models [🔗](#)

Pouyan Navard, Amin Monsefi, Mengxi Zhou, Wei-Lun Chao, Alper Yilmaz, Rajiv Ramnath

*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Workshop, 2025*

- Proposed a dual-pathway diffusion framework for controllable sketch-to-image generation with coarse-grained and fine-grained controllers linked by an adjustable knob mechanism, enabling high-quality synthesis from both novice and expert sketches.

### SeamCam: Quantifying Seamless Camouflage via Multi-Cue Visual Detectability

Amin Monsefi et al.

*European Conference on Computer Vision (ECCV), 2026 (under review)*

- Applied **Direct Preference Optimization (DPO) fine-tuning** to diffusion-based inpainting models for camouflage image generation; introduced a camouflage evaluation metric achieving 78.82% agreement with human judgments and a 1.5k-image benchmark.

### Large Language-and-Vision Assistant for Lunar Exploration (LLaVA-LE) [🔗](#)

Gokce Inal, Pouyan Navard, Alper Yilmaz

*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Workshop, 2026*

- Fine-tuned LLaVA for lunar surface characterization via two-stage instruction tuning, achieving a **3.3×** performance gain; curated LUCID, a 96k-image multimodal dataset with 81k QA pairs from NASA archives.

### **SegFormer3D: An Efficient Transformer for 3D Medical Image Segmentation** [↗](#)

Shehan Perera, **Pouyan Navard**, Alper Yilmaz

*IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Workshop, 2024*

- Designed a lightweight hierarchical transformer achieving competitive accuracy with **33×** **fewer** parameters and **13×** **lower** compute than state-of-the-art 3D segmentation models.

*Full publication list:* [Google Scholar](#) [↗](#)

## **Awards and Honors**

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### **Robert E. Altenhofen Memorial Scholarship Award** [↗](#)

*ISPRS, 2022*

- National-level scholarship recognizing excellence in photogrammetry and remote sensing research.

## **Education**

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### **The Ohio State University**

*Feb 2021 – Aug 2025*

*PhD in Electrical and Computer Engineering*

- GPA: 3.9/4.0
- Dissertation: *Distribution-Aware Learning for Sparsely Labeled and Imbalanced Spatiotemporal Data*
- Research focus on 3D perception, diffusion-based generative modeling, and multimodal learning
- Supervisor: Prof. Alper Yilmaz.

### **Isfahan University**

*Sep 2015 – Sep 2017*

*MSc in Electrical and Computer Engineering*

- GPA: 3.9/4.0
- Thesis: Geometric Structure from Motion under Large View Point Change using Epipolar Geometry

### **Isfahan University**

*Sep 2011 – Sep 2015*

*BSc in Electrical and Computer Engineering*

- GPA: 3.6/4.0
- Thesis: 3D View Photogrammetric Reconstruction